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Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Electronic Devices (EDC)

Paper Code: EC • Section 3: Electronic Devices (8-11 Marks)

Subject Code: EC-EDC • **Target Exam:** GATE ECE • **Weightage:** 8-11 Marks

1. Carrier Transport & Energy Band Theory

- **Intrinsic Concentration & Mass Action Law:** n_i^2 = N_c N_v e^{-E_g / (k_B T)}. In thermal equilibrium, n * p = n_i^2 holds for intrinsic, n-type, and p-type semiconductors.
- **Einstein Relation:** D_n / mu_n = D_p / mu_p = V_t = k_B T / q ≈ 25.86 mV at 300 K.
- **Current Density Equations:** Total current J = J_drift + J_diffusion.
  J_n = q n mu_n E + q D_n (dn/dx)
  J_p = q p mu_p E - q D_p (dp/dx)

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Mass Action Law	**n * p = n_i^2**	n, p = Electron, Hole Densities	Holds in thermal equilibrium regardless of donor/acceptor doping
Einstein Relation	**D / mu = V_t = k_B T / q ≈ 26 mV**	D = Diffusion, mu = Mobility	Connects thermal diffusion constant with field drift mobility
Built-in Potential	**V_{bi} = V_t * ln(N_A * N_D / n_i^2)**	N_A, N_D = Doping Levels	Contact potential across unbiased PN junction in equilibrium

GATE PYQ (2 Marks) - Diffusion Current Density
Question: A silicon bar doped with N_D = 10^16 cm^-3 has a hole gradient dp/dx = -10^19 cm^-4. If D_p = 12 cm^2/s, calculate hole diffusion current density J_p,diff.

Step 1: Formula: J_p,diff = -q * D_p * (dp/dx).
Step 2: Plug in values: q = 1.6 * 10^-19 C, D_p = 12 cm^2/s, dp/dx = -10^19 cm^-4.
Step 3: J_p,diff = - (1.6 * 10^-19) * 12 * (-10^19) = **+19.2 A/cm^2**.

2. MOSFET Physics: Threshold Voltage & Operating Modes

- **Operating Regions (NMOS):**
- *Cutoff:* V_GS < V_th => I_D = 0.
- *Triode (Linear):* V_GS >= V_th and V_DS < V_GS - V_th.
- *Saturation:* V_GS >= V_th and V_DS >= V_GS - V_th.
  I_D = (1/2) mu_n C_ox (W/L) (V_GS - V_th)^2 (1 + lambda V_DS).

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Saturation Condition	**V_{DS} >= V_{GS} - V_{th}**	V_th = Threshold Voltage	Pinch-off occurs at the drain end of channel
Saturation Current	**I_D = (1/2) mu_n C_{ox} (W/L) (V_{GS} - V_{th})^2**	W/L = Aspect Ratio	Square-law relationship ignoring channel-length modulation

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Transconductance g_m	**g_m = sqrt(2 mu_n C_{ox} (W/L) I_D)**	g_m = Transconductance	Small-signal voltage-to-current gain parameter

GATE PYQ (2 Marks) - NMOS Operating Region & Drain Current
Question: An NMOS has V_th = 0.7 V, mu_n C_ox (W/L) = 2.0 mA/V^2. If V_GS = 2.0 V and V_DS = 3.0 V, find the operating region and I_D.

Step 1: Overdrive voltage: V_ov = V_GS - V_th = 2.0 - 0.7 = 1.3 V.
Step 2: V_DS = 3.0 V > 1.3 V => **Saturation Region**.
Step 3: I_D = (1/2) * 2.0 mA/V^2 * (1.3 V)^2 = 1.0 * 1.69 = **1.69 mA**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** For one-sided step junctions (p+n), depletion width W and transition capacitance C_j are dictated almost entirely by the lighter-doped n-side.
- **Important:** Channel Length Modulation (lambda) results in an output resistance r_o = 1 / (lambda * I_D) in saturation.